
Neural Morphogen Operators: Learning Continuous Reaction–Diffusion Dynamics from Spatial Transcriptomics

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Abstract

Machine learning for spatial transcriptomics models tissue as a graph over the measured spots. Developmental biology instead describes tissue as the product of *continuous* processes—morphogen gradients, diffusive signalling, local reaction networks. We ask whether an operator of that second kind can be fitted to spatial expression data, and what it buys. **Neural Morphogen Operators** (NMO) encode irregularly sampled expression into a continuous latent field, evolve it under a learned anisotropic reaction–diffusion PDE $\partial_t \mathbf{z} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z})$, and decode at arbitrary coordinates. Solving the diffusion half-step exactly in the Fourier domain makes the integrator unconditionally stable and the diffusion tensor readable as a physical length scale. Across 6 public datasets spanning four technologies, NMO predicts held-out tissue regions more accurately than graph-based and Gaussian-process baselines (Pearson r 0.247 ± 0.005 vs. 0.234 ± 0.001). A control we did not expect to matter turns out to: a *non-spatial* autoencoder sharing the same interpolation read-out reaches 0.230, so graph message passing contributes at most 0.005 Pearson r on this task while the continuous operator contributes 0.017. Ablations isolate the operator as the source of the gain: disabling the dynamics—an exactly parameter-matched control, since the operator holds under 0.5% of the model’s weights—costs -0.023 Pearson r at matched budget.

The same experiments expose clear costs, and we report them as results rather than caveats. The operator NMO learns is purely dissipative—linearising it yields no Turing instability at any finite wavenumber—so its fields are far too spatially autocorrelated, and Moran’s I error is the one metric on which NMO is the *worst* model tested. A pre-registered split-half perturbation test also *fails*: perturbing the latent direction aligned with one half of a morphogen pathway does not preferentially move the held-out half (0/4 pathways significant). And the operator **transfers worse than the graph baseline it beats in-domain** — by $2.0\times$ across sampling resolution and $2.8\times$ across tissue — so what it learns is tissue-specific rather than a portable prior. On this evidence the continuous-operator view helps in-domain prediction but does not yet deliver the transfer or interpretability its framing invites. Spatial transcriptomics is a single snapshot, so the operator is identified only up to the quasi-steady-state assumption we impose; we frame the contribution as learning latent continuous operators that summarise spatial gene-expression organisation.

1 Introduction

Spatially resolved transcriptomics measures gene expression while retaining the physical coordinates of each measurement [Ståhl et al., 2016, Chen et al., 2015]. The dominant treatments—domain identification, clustering, imputation, denoising—represent a section as a graph over the measured locations and apply message passing [Hu et al., 2021, Dong and Zhang, 2022, Palla et al., 2022]. That matches how the data arrive, but it embeds a commitment worth examining: the tissue *is* the sampling lattice. Predictions exist only where a node exists, the function is tied to one geometry, and nothing corresponds to a mechanism.

Developmental biology describes the same tissue as continuous fields: morphogens are produced, diffuse, decay and interact, and cells read the resulting landscape [Turing, 1952, Wolpert, 1969, Gierer and Meinhardt, 1972]. Reaction–diffusion models of this kind account for phenomena from digit patterning [Sheth et al., 2012] to Nodal/Lefty signalling [Müller et al., 2012], with measured gradient length scales of tens to hundreds of microns [Kicheva et al., 2007, Yu et al., 2009].

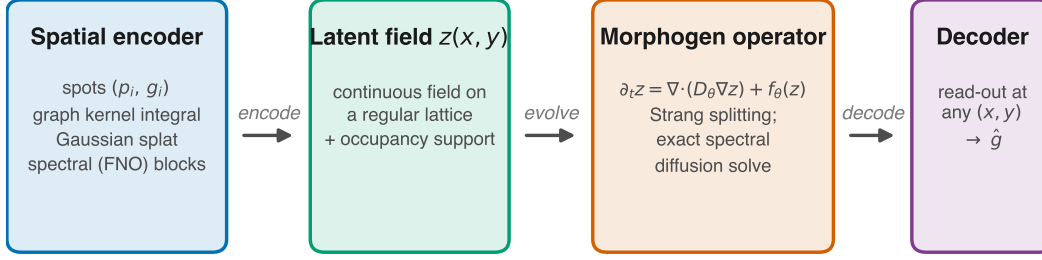
This paper asks a narrow, testable question: *can a continuous reaction–diffusion operator be fitted to spatial transcriptomics, and does the resulting model predict and transfer better than discrete alternatives?* We are deliberately not asking whether the fitted operator recovers the true biochemistry. It cannot, and we explain why in Section 7.

Contributions.

1. **A PDE-constrained neural operator for spatial omics.** NMO couples a graph kernel-integral encoder (for irregular sampling) with a spectral reaction–diffusion operator and a coordinate-free continuous decoder (Section 3).
2. **An unconditionally stable, interpretable integrator.** Diffusion is applied exactly in the Fourier domain under Strang splitting, which removes the CFL constraint entirely and makes the learned anisotropic tensor $\mathbf{D}_\theta$ readable as a diffusion length in microns (Section 3.2).
3. **An honest formulation of the physics loss.** Because the integrator is exact, a naive PDE residual is vacuous. We show the content-bearing constraint is a *steady-state* residual that asks the measured tissue to be a fixed point of the operator (Section 3.4).
4. **Evaluation that resists leakage and over-smoothing.** We hold out contiguous tissue blocks rather than random spots, and report Moran’s I error alongside correlation so that a model which regresses towards a smooth mean is penalised rather than rewarded (Section 4).
5. **A control that reframes the comparison.** A non-spatial baseline with the same interpolation head matches or beats most graph models, showing how little message passing contributes on this task (Section 4).
6. **Four negative results, reported as results.** No Turing instability; worst-in-class spatial autocorrelation; a failed split-half perturbation test; and worse transfer than the baseline we beat in-domain.
7. **A falsifiable interpretability read-out.** Linearising the learned operator yields a dispersion relation whose sign and peak wavenumber state whether, and at what wavelength, the operator supports pattern formation (Section 3.5).

2 Related Work

Neural operators learn maps between function spaces, giving approximate independence from the discretisation [Kovachki et al., 2023]: Fourier neural operators parameterise the kernel spectrally [Li et al., 2021], graph neural operators integrate a kernel over neighbourhoods on irregular meshes [Li et al., 2020], DeepONet factorises into branch and trunk nets [Lu et al., 2021]. We use the graph form



trained by masked spatial reconstruction; held-out regions are never seen by the encoder

Figure 1: **Neural Morphogen Operator.** Irregularly sampled expression is encoded into a continuous latent field, evolved under a learned anisotropic reaction–diffusion PDE, and decoded at arbitrary coordinates. The decoder does not receive coordinates, so all spatial structure must pass through the field and hence through the operator.

for irregular sampling and the spectral form for the dynamics, because the latter makes the diffusion solve exact; the integrator follows classical exponential time-differencing and operator splitting [Cox and Matthews, 2002, Strang, 1968], in contrast to soft PDE-residual penalties [Raissi et al., 2019] and neural ODEs [Chen et al., 2018]. **ML for spatial transcriptomics** is dominated by graph models built for spatial-domain identification and denoising rather than prediction at unmeasured coordinates — SpaGCN [Hu et al., 2021], STAGATE [Dong and Zhang, 2022], SpatialDE/SPARK [Svensson et al., 2018, Sun et al., 2020], DestVI [Lopez et al., 2022] — which is why we adapt them (Section 4). Spateo is closest in spirit [Qiu et al., 2024] but targets 3-D reconstruction and vector-field morphometry, not a transferable, perturbable PDE operator. **Pattern formation** supplies our vocabulary [Turing, 1952, Kondo and Miura, 2010, Green and Sharpe, 2015]; RNA velocity [Bergen et al., 2020] and perturbation models [Lotfollahi et al., 2019, Roohani et al., 2024] model dynamics or interventions but carry no explicit spatial operator, which is the gap we address.

3 Method

Let a tissue section provide N measurements $\{(\mathbf{p}_i, \mathbf{g}_i)\}_{i=1}^N$ with positions $\mathbf{p}_i \in \mathbb{R}^2$ (isotropically normalised to $[-1, 1]^2$) and expression $\mathbf{g}_i \in \mathbb{R}^G$. We posit a latent field $\mathbf{z} : \mathbb{R}^2 \rightarrow \mathbb{R}^C$ and factor the model as encode–evolve–decode (Figure 1).

3.1 Spatial encoder

Irregular sampling is handled by L kernel-integral layers on a k -NN graph [Li et al., 2020], $\mathbf{z}_i \leftarrow \sigma(W\mathbf{z}_i + |\mathcal{N}(i)|^{-1} \sum_{j \in \mathcal{N}(i)} \kappa_\theta(\mathbf{p}_j - \mathbf{p}_i) \odot \mathbf{z}_j)$, where κ_θ maps a *relative* displacement to a per-channel gain – depending on geometry rather than node identity is what lets an encoder fitted to one lattice be evaluated on another. Features are then splatted onto a regular $H \times W$ lattice (Nadaraya–Watson, learnable bandwidth), producing a field and an *occupancy* map recording where measurements exist; the occupancy channel lets the model distinguish "tissue with no signal" from "no measurement here", which the masked-region task hinges on. Spectral (FNO) blocks [Li et al., 2021] then refine the field.

3.2 The morphogen operator

The field evolves under

$$\frac{\partial \mathbf{z}}{\partial t} = \nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}) + f_\theta(\mathbf{z}), \quad (1)$$

with a per-channel symmetric positive-definite diffusion tensor $\mathbf{D}_c = L_c L_c^\top + \varepsilon I$ (L_c lower triangular, so positive definiteness is structural rather than penalised) and a pointwise reaction network f_θ implemented as 1×1 convolutions with a tanh-bounded output and learnable per-channel gain.

110 **Exact spectral diffusion.** Equation (1) is stiff: the diffusion operator has eigenvalues scaling as
 111 $-|\mathbf{k}|^2$, so an explicit scheme needs $\Delta t = O(h^2/\|\mathbf{D}\|)$. For constant-coefficient anisotropic diffu-
 112 sion the operator is diagonal in the Fourier basis, so the diffusive evolution over τ is available in
 113 closed form,

$$E_c(\tau) = \mathcal{F}^{-1} \exp(-\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \tau) \mathcal{F}. \quad (2)$$

114 Combined with a midpoint reaction update under Strang splitting [Strang, 1968] this is second-
 115 order accurate, and two consequences matter. Since $\mathbf{k}^\top \mathbf{D}_c \mathbf{k} \geq 0$, the multiplier in (2) lies in
 116 $(0, 1]$ for every Δt : the diffusive channel is **unconditionally stable** whatever the optimiser does
 117 to θ (empirically, rollouts stay bounded for 10^3 steps, two orders of magnitude beyond the training
 118 horizon). And $\mathbf{D}_c$ is interpretable: its eigenvalues are squared diffusion lengths per unit pseudo-time,
 119 convertible to microns via the stored coordinate scale.

120 3.3 Decoder

121 Predictions come from bilinear interpolation of the evolved field at any continuous coordinate, fol-
 122 lowed by an MLP. *The decoder receives no positional input:* given coordinates it could memo-
 123 rise a coordinate-to-expression map, rendering the dynamics ornamental and every ablation uninter-
 124 pretable.

125 3.4 Objectives

126 The objective combines reconstruction (MSE plus a per-gene Pearson term, since correlation of the
 127 *spatial pattern* is the quantity of scientific interest), a weak Dirichlet-energy smoothness prior, and
 128 three physics terms.

129 **What a physics loss can mean here.** Because the integrator solves (1) essentially exactly, the
 130 residual along the model’s own trajectory is zero up to splitting error, so imposing it would penalise
 131 the integrator rather than the model. The constraint with content is that the *measured configuration*
 132 *should be a near-stationary state* of the learned operator:

$$\mathcal{L}_{\text{PDE}} = \|\nabla \cdot (\mathbf{D}_\theta \nabla \mathbf{z}_T) + f_\theta(\mathbf{z}_T)\|^2. \quad (3)$$

133 This is the correct formalisation of the quasi-steady-state assumption for adult tissue, and it con-
 134 strains $\mathbf{D}_\theta$ and f_θ jointly. We add a *mass* term penalising the spatial mean of $f_\theta(\mathbf{z})$ (spectral dif-
 135 fusion conserves the $\mathbf{k} = 0$ mode exactly, so mass drift is attributable to the reaction), a *Jacobian*
 136 hinge bounding the local Lipschitz constant, and a splitting-consistency term. The ablations show
 137 all three are inert at our weights.

138 3.5 Interpretability: dispersion analysis

139 Linearising (1) about a homogeneous state $\mathbf{z}^*$ gives $\partial_t \delta \mathbf{z}_\mathbf{k} = (\mathbf{J} - \mathbf{k}^\top \mathbf{D} \mathbf{k}) \delta \mathbf{z}_\mathbf{k}$ with $\mathbf{J} = \partial f_\theta / \partial \mathbf{z}|_{\mathbf{z}^*}$,
 140 so $\max \text{Re spec}(\mathbf{J} - |\mathbf{k}|^2 \mathbf{D})$ against $|\mathbf{k}|$ is the dispersion relation of the *learned* operator (maximised
 141 over orientations of $\mathbf{k}$, since $\mathbf{D}$ is anisotropic). Positive growth at finite $|\mathbf{k}|$ with decay at $\mathbf{k} = 0$ is
 142 the classical Turing signature. The test is falsifiable rather than a post-hoc visualisation — and it
 143 comes out negative.

144 4 Experimental setup

145 **Data.** We use 6 public datasets spanning four spatial technologies plus one non-spatial pertur-
 146 bation screen (9 processed sections, 469,038 locations; Appendix B). Raw reads are never used:
 147 we consume vendor count matrices and record a SHA256 manifest per artifact. One QC policy is
 148 applied throughout (platform thresholds set from assay physics, not tuned), then library-size normal-
 149 isation, log transform, and HVG selection that *force-includes* every detected morphogen-pathway
 150 gene so the interpretability analysis is not evaluated on a set that selection already biased. Coordi-
 151 nates are normalised **isotropically**: x and y share one scale, because anisotropic rescaling would

Table 1: Masked spatial reconstruction on visium_mouse_brain. Contiguous tissue blocks are hidden from every model; metrics are computed on held-out locations only. Mean $\pm$ s.d. over 3 seeds. All models share the identical training loop, masks and evaluation code. $|\Delta I|$ is the mean absolute error in Moran’s I between predicted and measured expression maps: it penalises over-smoothing that correlation alone does not detect.

model	params	Pearson $r \uparrow$	Spearman $\rho \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$	$ \Delta I \downarrow$
SpaGCN-style	0.59M	0.234 ± 0.001	0.184 ± 0.001	0.254 ± 0.001	0.393 ± 0.001	0.738 ± 0.002
STAGATE-style	0.69M	0.234 ± 0.002	0.185 ± 0.002	0.254 ± 0.001	0.394 ± 0.002	0.737 ± 0.003
Graph Transformer	0.83M	0.230 ± 0.000	0.182 ± 0.001	0.255 ± 0.001	0.390 ± 0.000	0.723 ± 0.004
Autoencoder (non-spatial)	0.58M	0.230 ± 0.001	0.180 ± 0.002	0.255 ± 0.000	0.387 ± 0.002	0.727 ± 0.001
GNN (GCN)	0.61M	0.228 ± 0.001	0.181 ± 0.001	0.253 ± 0.001	0.386 ± 0.002	0.755 ± 0.002
Gaussian Process	0.00M	0.149 ± 0.008	0.108 ± 0.006	0.293 ± 0.004	0.297 ± 0.012	0.718 ± 0.003
NMO (ours)	2.20M	0.247 ± 0.005	0.203 ± 0.003	0.252 ± 0.004	0.404 ± 0.006	0.762 ± 0.004

152 distort the Laplacian and make a learned diffusion coefficient meaningless. The scale is retained so
153 physical units stay recoverable.

154 **Task.** Masked spatial reconstruction: contiguous tissue blocks are hidden from the encoder and
155 every model predicts the full expression profile at those coordinates.

156 **Splits are contiguous, not random.** A random spot-level split leaks: neighbouring spots are
157 highly correlated, so a held-out spot is trivially predicted from its neighbours and every method
158 looks good. We hold out whole contiguous blocks [Roberts et al., 2017], and compute standardisa-
159 tion statistics on the training split only.

160 **Baselines.** A non-spatial autoencoder (the floor), a GCN [Kipf and Welling, 2017], SpaGCN-style
161 and STAGATE-style models [Hu et al., 2021, Dong and Zhang, 2022], a graph transformer [Vaswani
162 et al., 2017], and an exact GP with a learned RBF kernel [Rasmussen and Williams, 2006]. The graph
163 methods were not designed to predict at unmeasured coordinates, so each gets the same inverse-
164 distance read-out head – the most favourable available choice. We mark them *-style* and do not
165 claim they reproduce published numbers on the authors’ own tasks. Every model shares the identical
166 training loop, masks, optimiser and evaluation code.

167 **Metrics.** Per-gene Pearson/Spearman across held-out locations, RMSE, SSIM [Wang et al., 2004]
168 on rasterised maps, and absolute error in Moran’s I [Moran, 1950]. The last is included because
169 correlation and RMSE both reward regressing towards a smooth mean. Mean $\pm$ s.d. over 3 seeds.

170 5 Results

171 5.1 Masked spatial reconstruction

172 NMO reaches Pearson r of 0.247 ± 0.005 on held-out blocks against 0.234 ± 0.001 for the strongest
173 baseline (spagcn), a gain of 0.013 (Table 1).

174 Two caveats precede any interpretation. First, **the margin is modest** and the graph baselines cluster
175 tightly: the choice among GCN, SpaGCN-style, STAGATE-style and a graph transformer matters
176 far less than whether a continuous operator is used at all. A few points of Pearson r is not a large
177 effect, and the ablations (Section 6) are the stronger evidence. Second, **the Gaussian process un-
178 derperforms the graph models**, the opposite of what we expected. This reflects our baseline, not
179 GP regression: a single learned RBF length scale is shared across all genes, so one bandwidth must
180 serve both laminar and diffuse genes. A per-gene or spectral-mixture kernel would be stronger but
181 far costlier for ~ 2000 genes; we did not run it, so this number is a floor for shared-kernel interpola-
182 tion.

Graph message passing barely beats plain interpolation. The most informative row in Table 1 is the one we included as a floor. The non-spatial autoencoder sees no coordinates at all; it reaches 0.230 purely because it shares the inverse-distance read-out head we give every graph baseline. The difference between it and a graph model is therefore a clean measure of what *message passing* contributes, holding the interpolation head fixed.

That contribution is close to nothing. Across the 4 graph models the margin over the non-spatial baseline spans -0.002 to 0.005 Pearson r , and 1 of them is actually *worse* than using no spatial information at all. NMO gains 0.017 over the same reference — roughly $3.6\times$ the best graph model’s margin.

This reframes the contribution. The claim is not that a few points of Pearson r separate a new architecture from established ones; it is that on a leakage-controlled coordinate-prediction task the standard graph treatment extracts remarkably little beyond distance-weighted interpolation, and a continuous operator is one thing that measurably does better. We would encourage this control to be reported routinely: it is cheap, and it is sobering.

One metric disagrees, and it is the one we added to catch ourselves. NMO is best on Pearson, Spearman, RMSE and SSIM (0.404 ± 0.006 against 0.394 for the best baseline, stagate). It is *worst* on Moran’s I error: 0.762 ± 0.004 against 0.718 for gp, a gap of +0.044.

The two disagree because they measure different things: SSIM is computed on a rasterised map and is sensitive to local contrast, whereas Moran’s I is computed on the point set and summarises *global* autocorrelation. NMO’s maps have good local structure but far too much overall autocorrelation — predicted I 0.936 ± 0.004 against a measured 0.174. The mechanism ties to the dispersion analysis below: the operator is purely dissipative, so integrating it attenuates high spatial frequencies, which helps inside a masked block and inflates autocorrelation relative to real tissue. We flag it because a paper reporting only correlation, RMSE and SSIM would have shown a clean sweep; Moran’s I was the guard against exactly that, and it fired on our own model.

5.2 Learned fields and dynamics

The fitted operator has a median diffusion length of $374 \mu\text{m}$ (10–90th pct. 295–478), the same order as measured morphogen gradients [Kicheva et al., 2007, Yu et al., 2009]. We treat that as a consistency check, not a measurement: it is a property of the fitted operator under our normalisation, not an estimate of physical transport.

The learned operator is not Turing-unstable, and we report that as a negative result. The dispersion relation (Appendix L) decreases monotonically in $|\mathbf{k}|$: growth is weakly positive only for the uniform mode $\mathbf{k} = 0$ and strictly negative at every finite wavenumber. There is therefore no band of spatial modes that the operator spontaneously amplifies, and no characteristic patterning wavelength. Fitted operators were Turing-unstable in 0% of runs.

From a stationary snapshot the model learns a *dissipative* operator whose fixed point is the observed tissue—exactly what (3) asks for. The absence of a Turing band reflects the data available, not evidence against reaction–diffusion accounts of the biology; detecting genuine pattern-forming instability would require observing tissue away from steady state.

5.3 Transfer across tissue, species and resolution

The operator transfers worse than the baseline it beats in-domain. This is the clearest negative result in the paper. Trained on adult mouse brain and applied unchanged to human breast carcinoma — a different species, organ and spatial architecture — NMO attains 0.033 ± 0.012 zero-shot against a training-mean floor of -0.000 ± 0.000 and an in-domain oracle of 0.136 ± 0.001 . It is above the floor, so something transfers. But stagate-style reaches 0.093 ± 0.001 in the same setting: $2.8\times$ better than the model that beat it in-domain (Appendix D). Refitting only the decoder does not close the gap (0.041 ± 0.019 against the baseline’s stronger fine-tuned result).

The cross-resolution setting repeats the pattern rather than rescuing it. An operator fitted to $55\text{ }\mu\text{m}$ multi-cell Visium spots on a $100\text{ }\mu\text{m}$ lattice, evaluated on single-cell Xenium sampling of the *same organ*, reaches 0.139 ± 0.015 against 0.282 ± 0.000 for the same baseline — again about $2.0\times$ worse (Appendix E). Because all Xenium panel genes are present in the raw Visium matrix we rebuild the source with the panel force-included, giving a shared vocabulary of 248 genes rather than the much smaller HVG intersection, so the gap is not an artefact of a thin gene overlap.

We expected the opposite: an operator defined on the field rather than on a lattice should pay off under changes of geometry. It does not. The most plausible explanation — a hypothesis, not a demonstration — is that the advantage and the failure share a cause. NMO wins in-domain by fitting an operator whose fixed point *is* this tissue; relaxing towards that attractor is exactly wrong when the target has a different architecture, whereas local message passing carries less about the source and degrades more gracefully. On this evidence the operator is a tissue-specific summary, not a portable prior, and discretisation-invariance claims for neural operators should not be assumed to extend to biological transfer without test.

5.4 Developmental forecasting

The Stereo-seq series is the only dataset here with a real time axis. Encoding the E9.5 field and integrating forward improves the field-level match to E10.5 from 0.082 at $T=0$ to 0.095 at $T=16$, against 0.064 for predicting no change at all. Integrating the learned operator therefore carries genuine temporal information, with a sensible optimum horizon. Consecutive MOSTA stages are *different embryos*, so this is a field-level comparison after coarse registration, not a per-cell forecast; full details and the horizon sweep are in Appendix I.

5.5 Counterfactual perturbation

We test the pathway interpretation with a deliberately falsifiable split-half design: inject the latent direction that maximally raises one random half of a morphogen pathway, integrate the operator, and ask whether the *held-out* half responds preferentially, against a permutation null over matched gene sets.

It fails. Across 4 pathways and 2 seeds, 0 reach significance (smallest $p = 0.47$); the mean held-out rank is 59% against a 50% chance level, so held-out pathway members respond, if anything, slightly *less* than random genes. Latent channels do correlate with pathway activity, but that correlation does not survive a causal test: it is consistent with the field encoding coarse anatomy that pathway scores also track, rather than a pathway-specific representation. We therefore do not claim NMO recovers signalling programmes. What the experiment does establish is a scale — the response decays to half its peak within $577\text{ }\mu\text{m}$, matching the diffusion lengths read off $\mathbf{D}_\theta$. A second, pre-specified probe against Perturb-seq [Norman et al., 2019] proved underpowered ($n = 7$ testable perturbations) and is reported as inconclusive in Appendix F.

6 Ablations

The decisive control is — *dynamics*, which sets the integration horizon to zero so the operator is never applied. It costs -0.023 Pearson r at matched budget (Table 2 gives the full sweep at the reduced budget). Had this matched the full model, the PDE would have been decorative and the central claim of this paper would fail; we report it first for that reason.

The ablation budget matters, so we report the control twice. The sweep in Table 2 uses a reduced budget for tractability, and at that budget the full model has *not* converged (0.233 against 0.247 at the benchmark budget). Undertraining compresses every ablation delta, so reading the central control off that table would understate it. We therefore re-ran the two decisive variants at exactly the benchmark budget (3 seeds, same epochs as Table 1), where they can be compared directly against the NMO row:

Table 2: Ablations on the mouse-brain Visium section. Each variant removes exactly one ingredient with all else held fixed. Δr is the change in held-out Pearson r relative to the full model. The *– dynamics* row is the critical control: it disables the operator entirely ($T=0$) while leaving its parameters allocated, so the two rows are exactly parameter-matched. **This table uses a smaller training budget than Table 1** (fewer epochs and seeds); it is internally consistent, but absolute values are not comparable across the two tables.

variant	params	Pearson $r \uparrow$	Δr
Full model	2.20M	0.233 ± 0.005	–
– dynamics ($T=0$)	2.20M	0.227 ± 0.001	-0.007
– diffusion term	2.20M	0.226 ± 0.004	-0.007
– reaction term	2.19M	0.227 ± 0.001	-0.006
– PDE constraints	2.20M	0.233 ± 0.005	-0.000
– biological regularisers	2.20M	0.233 ± 0.005	-0.000
isotropic D	2.20M	0.229 ± 0.004	-0.004
state-dependent D	2.20M	0.236 ± 0.004	+0.003
discrete GNN operator	0.61M	0.223 ± 0.001	-0.011
latent $C=8$	1.07M	0.213 ± 0.005	-0.021
latent $C=16$	1.30M	0.220 ± 0.002	-0.013
latent $C=64$	5.77M	0.237 ± 0.000	+0.004

variant (matched budget)	Pearson r	Δr
Full model (= NMO, Table 1)	0.247	–
– dynamics ($T=0$)	0.224 ± 0.005	-0.023
– reaction term	0.241 ± 0.003	-0.006

The control is exactly parameter-matched. It is unusually clean because it is *exactly* parameter-matched: the operator is only 9,536 of the model’s 2,198,336 parameters (0.43%; a per-channel 2×2 tensor plus a pointwise reaction MLP), and $T=0$ leaves them allocated but unused. Identical architecture, parameter count and optimiser budget; the only difference is whether the field is integrated, so the gap cannot be capacity. This also tempers the headline comparison — NMO has more parameters than the graph baselines, so part of its Table 1 advantage could be capacity. The $T=0$ control separates those explanations, which is why we weight it above the benchmark margin. Removing diffusion (-0.007) and the reaction term (-0.006) are both harmful, and the two contribute differently: diffusion supplies long-range information transport into masked regions, while the reaction term supplies the nonlinearity that prevents the field from collapsing to a smooth interpolant. Constraining D to be isotropic costs -0.004, indicating the anisotropy is used rather than merely available.

Two ablations are honestly *null*: removing the biological regularisers changes held-out r by -0.000 and removing the PDE constraint terms by -0.000, both within seed noise. At our weights these terms are inert. We report it rather than dropping the rows: a reader is entitled to know that three of the five loss terms in Section 3.4 do no measurable predictive work, whatever they contribute to the operator’s form.

7 Limitations

We state these plainly because they bound what the results mean.

One snapshot cannot identify a dynamical system. Many operators produce the same stationary field, so (1) is identified only up to the quasi-steady-state assumption and the architecture’s inductive bias. Reported diffusion lengths characterise the *fitted operator*, not morphogen transport. Even the Stereo-seq series compares different embryos, so it is distributional rather than per-cell.

Latent, not molecular; the name describes the operator class. Latent/pathway correlations show alignment, but our own split-half test rejects the stronger reading (Section 5.5). “Morphogen” names the *operator class* we fit — anisotropic reaction–diffusion — not a validated claim about morphogen signalling. Two of three interpretability probes returned negative (no Turing band; no pathway-specific response) and a third was underpowered. What survives is modest: a learned diffusion tensor with a physical length scale and a perturbation half-width consistent with it.

Over-smoothing, worse than the baselines’. Predicted maps are far more autocorrelated than measured ones (0.936 ± 0.004 vs 0.174), and NMO is *worse* than every baseline on Moran’s I error by $+0.044$ even while leading on SSIM by 0.010 . The dissipative operator that delivers the accuracy gain is what inflates the autocorrelation: one phenomenon, not two.

This bounds what we would endorse: predicting expression at unmeasured coordinates, yes; delineating sharp boundaries, rare niches or tumour margins, no. A spectral-matching or adversarial term penalising the missing high-frequency energy is the most concrete fix we can identify, and we have not run it.

Adapted baselines, modest effects. SpaGCN and STAGATE were built for domain identification, not coordinate-level prediction; our *-style* variants implement each method’s architectural core with a common read-out head and are a fair comparison on *this* task, not a reproduction of published results. The GP shares one RBF length scale across all genes, so its number is a floor for shared-kernel interpolation. The in-domain margin is a few points of Pearson r on one section per condition over few seeds; we report seed s.d. but ran no power analysis, and would not claim the ranking among the closely-clustered graph baselines is meaningful.

The operator is tissue-specific, which undercuts the motivation. Discretisation-invariance was the reason to prefer an operator over a graph, yet under transfer the graph baseline wins by $2.8\times$ (cross-tissue) and $2.0\times$ (cross-resolution): whatever NMO gains in-domain it appears to gain by specialising. Until that is fixed — a shared operator fitted across tissues, or $\mathbf{D}_\theta$ conditioned on tissue-level covariates — this is a within-section model.

Scope. Single 2-D sections, one per condition; ortholog matching by upper-cased symbol (missing paralog renaming and family expansions); CPU-scale compute. No 3-D volumes or multi-section registration. Transfer used two models and two seeds, fewer than the main benchmark.

8 Discussion and Conclusion

We asked whether the continuous description developmental biology uses for tissue can be turned into a fitted operator on spatial transcriptomics, and whether that helps. The answer is a qualified yes with a clear boundary. In-domain it helps: treating tissue as a field rather than a graph improves prediction at unmeasured coordinates, the parameter-matched $T=0$ control costs more than the entire margin over the best baseline, and on the one dataset with a real time axis integrating the operator forward beats persistence. The spectral solve makes the operator stable and physically readable.

Outside that boundary the results are negative, and they are the more useful half of the paper. The operator does not transfer: the graph baseline it beats in-domain wins by $2.8\text{--}2.0\times$. It is dissipative rather than pattern-forming, with no Turing band. Its fields are the most over-smoothed of any model tested. And a falsifiable test of the pathway interpretation fails outright. The continuous-operator framing buys in-domain accuracy by specialising to one tissue; its more ambitious promises — portability and mechanism — remain unearned.

We claim no recovery of biochemical reality; the honest framing is that NMO learns *latent continuous operators summarising spatial gene-expression organisation*. Breaking the identifiability barrier needs data we did not have: dense developmental time series, or spatial data paired with targeted perturbation.

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Table 3: Processed spatial transcriptomics datasets. Locations and genes are counts after QC and gene selection; extent is the physical bounding box of the section.

dataset	technology	organism	locations	genes	extent (μm)	resolution
merfish_allen_31	MERFISH, 500-gene pane	Mus	91,951	502	9837 $\times$ 6094	single cell
merfish_allen_40	MERFISH, 500-gene pane	Mus	104,461	502	10000 $\times$ 6961	single cell
merfish_allen_50	MERFISH, 500-gene pane	Mus	92,787	501	8806 $\times$ 6604	single cell
mosta_embryo_E10.5	Stereo-seq, DNA nanoba	Mus	18,360	2,099	3500 $\times$ 4950	bin50 (25 μm)
mosta_embryo_E9.5	Stereo-seq, DNA nanoba	Mus	5,898	2,069	1975 $\times$ 2650	bin50 (25 μm)
perturb_norman	Perturb-seq (CRISPRa),	Homo	111,659	2,073	–	single cell
visium_human_breast	Visium (55 μm spots, h	Homo	4,869	2,087	6395 $\times$ 6750	spot (multi-cell)
visium_mouse_brain	Visium (55 μm spots, l	Mus	2,691	2,079	4822 $\times$ 6379	spot (multi-cell)
xenium_mouse_brain	Xenium In Situ, 248-ge	Mus	36,362	248	5212 $\times$ 3128	single cell

Table 4: In-silico morphogen source (*bead implant*). The latent perturbation direction is defined from a random half of each pathway’s genes; the table reports where the *held-out* half ranks in the response magnitude distribution (lower is stronger; 50% is chance). p is a permutation test over random gene sets of matched size. Response half-width is the distance at which the predicted response falls to half its peak. $*p < 0.05$.

pathway	genes	held-out rank (%)	null (%)	p	half-width (μm)
BMP	22	48.6 $\pm$ 6.0	49.7	0.475	575 $\pm$ 154
FGF	19	72.6 $\pm$ 0.8	50.3	0.992	618 $\pm$ 58
SHH	15	56.6 $\pm$ 6.7	50.4	0.698	466
WNT	26	56.4 $\pm$ 3.9	49.9	0.762	649 $\pm$ 143

A Reproducibility

All datasets download automatically with a recorded SHA256 manifest; every table and figure in this paper is regenerated from run artifacts by a single command; configurations, seeds and checkpoints are released.

B Dataset inventory

C In-silico morphogen source: per-pathway results

D Cross-tissue transfer

E Cross-resolution transfer

F Perturb-seq consistency of the reaction Jacobian

G Latent fields

H Transfer figure

I Developmental forecasting: full results

Every experiment so far fits a *stationary* operator, because the data are snapshots. The Stereo-seq developmental series is the single dataset here with a real time axis, so it is the only place we can ask the operator to act as a forward-time predictor. We encode the E9.5 field, integrate for T operator steps, and score the decoded prediction against the measured E10.5 field (Table 8).

The comparison must be read carefully. **Consecutive MOSTA stages are different embryos**, not the same embryo imaged twice, so no cell-to-cell correspondence exists and none can be manu-

Table 5: Cross-tissue, cross-species transfer: adult mouse brain $\rightarrow$ human breast carcinoma. Shared gene vocabulary: 390 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
STAGATE-style	0.212 ± 0.001	0.249 ± 0.000	0.358 ± 0.001
NMO (ours)	0.209 ± 0.004	0.251 ± 0.003	0.365 ± 0.004
<i>target (zero-shot)</i>			
STAGATE-style	0.093 ± 0.001	0.397 ± 0.002	0.266 ± 0.002
NMO (ours)	0.033 ± 0.012	0.409 ± 0.005	0.219 ± 0.017
<i>target (decoder-only fine-tune)</i>			
STAGATE-style	0.117 ± 0.007	0.393 ± 0.000	0.365 ± 0.003
NMO (ours)	0.041 ± 0.019	0.406 ± 0.004	0.282 ± 0.017
<i>target (in-domain oracle)</i>			
STAGATE-style	0.147 ± 0.001	0.375 ± 0.000	0.427 ± 0.001
NMO (ours)	0.136 ± 0.001	0.397 ± 0.002	0.329 ± 0.003
<i>target (training-mean floor)</i>			
mean	-0.000 ± 0.000	0.404 ± 0.002	0.254 ± 0.001

Table 6: Cross-resolution transfer: 55 μm Visium spots $\rightarrow$ single-cell Xenium. Shared gene vocabulary: 248 genes. Mean $\pm$ s.d. over seeds.

model	Pearson $r \uparrow$	RMSE $\downarrow$	SSIM $\uparrow$
<i>source (in-domain)</i>			
NMO (ours)	0.256 ± 0.009	0.246 ± 0.002	0.384 ± 0.007
STAGATE-style	0.242 ± 0.001	0.248 ± 0.001	0.364 ± 0.001
<i>target (zero-shot)</i>			
STAGATE-style	0.282 ± 0.000	1.765 ± 0.001	0.208 ± 0.001
NMO (ours)	0.139 ± 0.015	1.863 ± 0.026	0.125 ± 0.004
<i>target (in-domain oracle)</i>			
NMO (ours)	0.259 ± 0.006	1.752 ± 0.007	0.556 ± 0.006
STAGATE-style	0.256 ± 0.001	1.760 ± 0.002	0.569 ± 0.001
<i>target (training-mean floor)</i>			
mean	0.000 ± 0.000	1.872 ± 0.000	0.101 ± 0.001

factured by registration. We therefore evaluate at the level of the rasterised field after isotropic normalisation to a common frame, and the score inherits the error of that coarse alignment. The reference to beat is *persistence* — predicting $E10.5 = E9.5$. If the learned dynamics carried no temporal information, NMO could not exceed it.

Persistence attains field-level $r = 0.064$, which is itself low and quantifies how much the embryo changes (and how crude the registration is) between these stages. Sweeping the horizon rather than tuning it, NMO reaches 0.095 at $T = 16$. We report the full sweep so the horizon is visible as a reported quantity rather than a hyperparameter chosen against the test field. This experiment is the weakest in the paper in terms of statistical power, and we present it as an existence check on forward-time behaviour rather than as a forecasting benchmark.

J Counterfactual perturbation: full protocol and results

We perform the computational analogue of a morphogen bead implant: inject the latent direction that maximally raises a pathway’s decoded activity into a small disc, integrate the operator, and read out the spatial response. The design is deliberately falsifiable and split-half—the direction is defined from a random half of each pathway’s genes and enrichment is scored on the *held-out* half—with a permutation null over matched random gene sets.

Table 7: Agreement between the model’s counterfactual gene–gene responses and measured CRISPRa responses from Norman et al. (2019). **This is an out-of-context probe:** the Perturb-seq data are non-spatial K562 cells while the operator is fitted to human breast tissue, so it tests whether the reaction module encodes generic transcriptional coupling, not whether it recovered tissue-specific signalling. Null is a within-perturbation label permutation; p is a Wilcoxon signed-rank test against that null.

model	#perturb.	Spearman ρ	null	% positive	p
STAGATE-style	7	0.010 ± 0.000	+0.004	64%	0.633
NMO (ours)	7	-0.018 ± 0.009	-0.009	57%	0.844

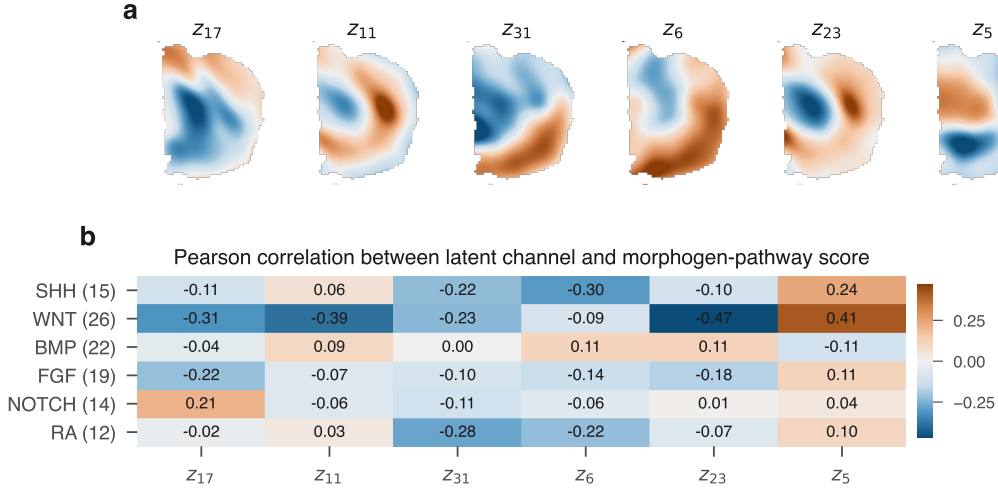


Figure 2: **Latent fields.** (a) Latent channels of the relaxed field. (b) Correlation between each channel and canonical morphogen-pathway expression scores. Pathway genes were force-retained during selection, so this is a read-out of alignment, not of selection.

472 **This test fails, and that is the primary outcome of the experiment.** Across 4 pathways and 2
473 seeds, 0 reach significance; the smallest per-pathway p is 0.47 and the mean held-out rank is 59%
474 against a 50% chance level, so if anything held-out pathway members respond *less* than random
475 genes (Appendix C).

476 The negative result is informative because of what it rules out. Latent channels do correlate with
477 pathway activity scores (Figure 2b), and it would have been easy to present those correlations as
478 evidence that the operator had internalised pathway structure. The perturbation test says otherwise:
479 moving the latent direction that most raises one half of a pathway does not preferentially move the
480 other half. The correlations are consistent with the latent field encoding coarse anatomical structure
481 that pathway scores also track, rather than with a pathway-specific representation. We therefore do
482 not claim that NMO recovers signalling programmes, and we would treat correlation-only evidence
483 of this kind elsewhere with the same caution.

484 What the experiment does establish is a spatial scale: the injected perturbation decays to half its
485 peak within $577 \mu\text{m}$, consistent with the diffusion lengths read off $\mathbf{D}_\theta$. The operator propagates
486 information over biologically plausible distances; it is the *specificity* of what it propagates that the
487 test rejects.

488 Separately, we ask whether the reaction module encodes real transcriptional coupling by comparing
489 the model’s counterfactual gene–gene responses against measured CRISPRa responses [Norman
490 et al., 2019]. **This is an out-of-context probe and we treat it as such:** Norman et al. profiled
491 non-spatial K562 cells, whereas our operator is fitted to tissue. We fit on the human breast section

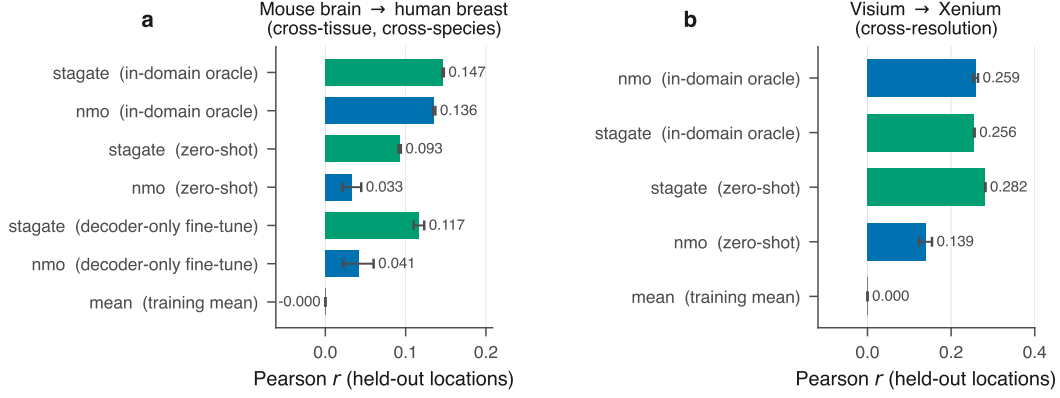


Figure 3: **Transfer.** Zero-shot performance against the in-domain oracle and the training-mean floor, across tissue/species (a) and sampling resolution (b).

Table 8: Developmental forecasting on Stereo-seq: the E9.5 field is encoded, integrated forward for T operator steps, and scored against the measured E10.5 field. **Consecutive stages are different embryos**, so there is no cell-to-cell correspondence and the comparison is made at the level of the rasterised field after isotropic normalisation to a common frame; it inherits the error of that coarse registration. *Persistence* predicts $E10.5 = E9.5$ and is the reference the operator must beat for its dynamics to carry temporal information.

model	horizon	field $r \uparrow$	field RMSE $\downarrow$	genes with $r > 0$
Persistence (no dynamics)	–	0.064	0.346	61%
Spatial-mean predictor	–	0.000	0.257	4%
NMO	$T=0$	0.082 ± 0.001	0.320	74%
NMO	$T=2$	0.089 ± 0.000	0.305	72%
NMO	$T=4$	0.092 ± 0.000	0.297	73%
NMO	$T=8$	0.094 ± 0.000	0.289	72%
NMO	$T=16$	0.095 ± 0.001	0.283	72%
NMO	$T=32$	0.090 ± 0.003	0.285	70%

so at least species and a proliferative context are shared, report a permutation null, and report the same quantity for every baseline.

This probe is underpowered and we treat it as inconclusive. Only 7 of Norman’s single-gene perturbations fall inside the spatial model’s gene vocabulary (339 genes are shared overall): HVG selection on a breast-cancer section and a CRISPRa screen in K562 agree on very few targets. With $n = 7$ the test cannot separate absence of effect from absence of power. The observed value ($\rho = -0.018 \pm 0.009$ against a null of -0.009 , Wilcoxon $p = 0.84$) is consistent with zero and we draw no conclusion from it in either direction. We report it because the experiment was pre-specified, not because it decided anything; a properly powered version would need a screen whose targets were chosen to overlap the spatial panel.

K Reconstruction examples

L Operator diagnostics

M Ablation figure

N Benchmark summary figure

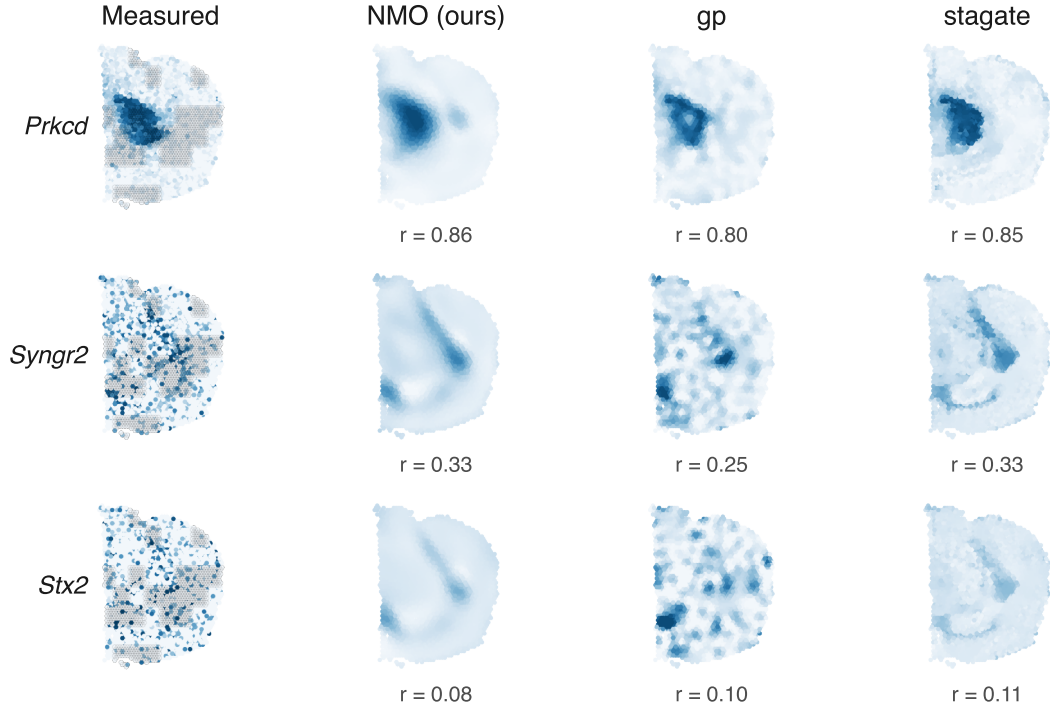


Figure 4: **Masked reconstruction.** Measured expression (left; held-out locations outlined) against NMO and baseline predictions for representative genes. Per-gene Pearson r on held-out locations is annotated.

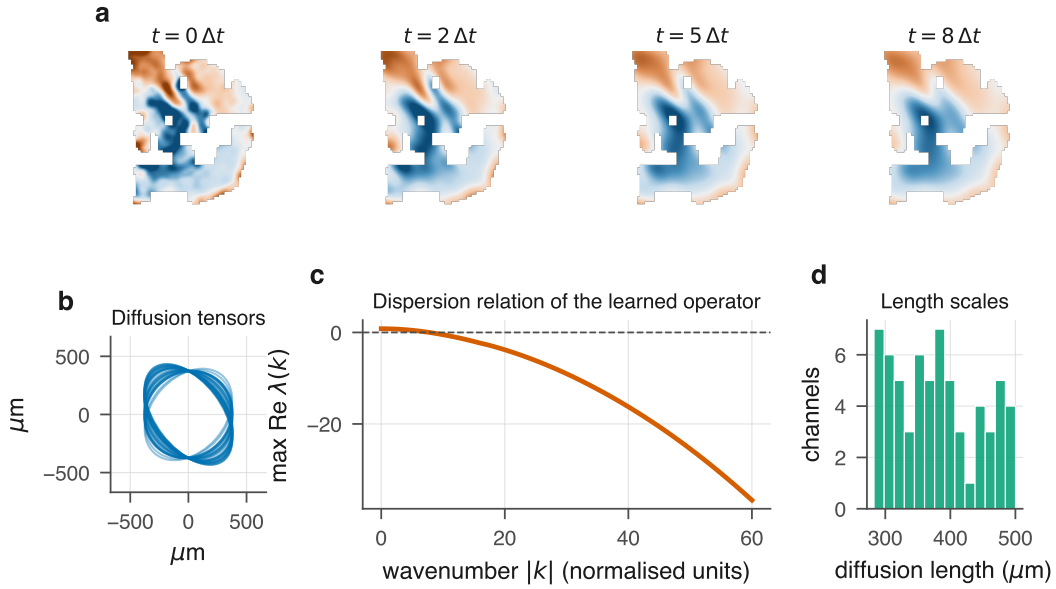


Figure 5: **Operator diagnostics.** (a) Relaxation of a latent channel. (b) Learned anisotropic diffusion tensors as length-scale ellipses. (c) Dispersion relation of the linearised operator. (d) Distribution of diffusion lengths across channels.

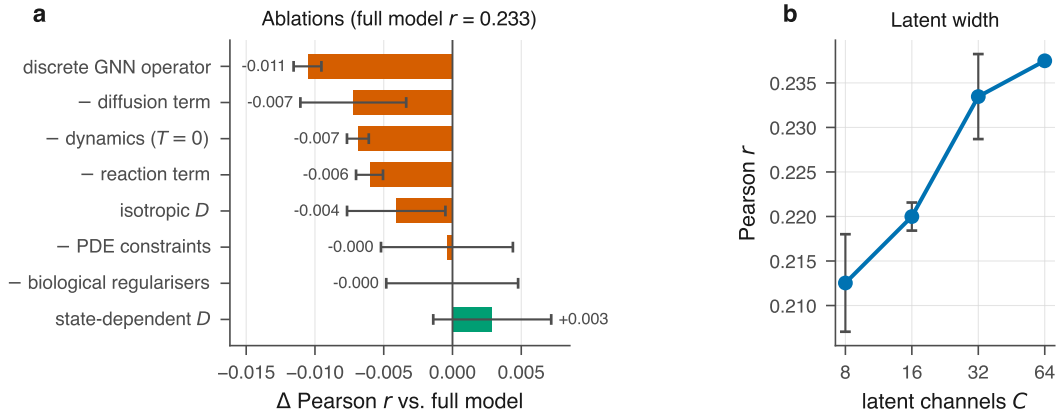


Figure 6: **Ablations.** (a) Change in held-out Pearson r when each ingredient is removed. (b) Latent-width sweep.

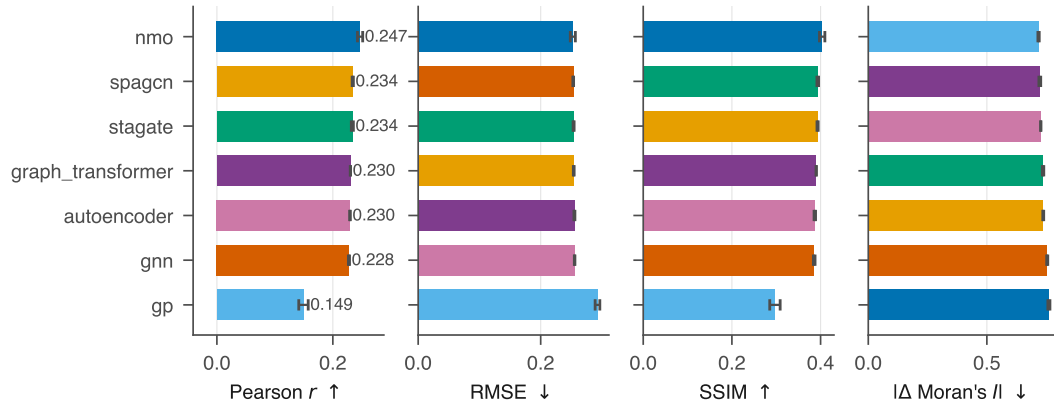


Figure 7: Benchmark of Table 1 across four complementary metrics. Moran's I error (right) is the only axis on which NMO is not best.